P99 CONF

A ScyllaDB Community

PostgreSQL Goes Columnar: How Far Can It Really Push Analytics?



Daniel Seybold
Co-Founder

benchANT

Daniel Seybold (he/him)

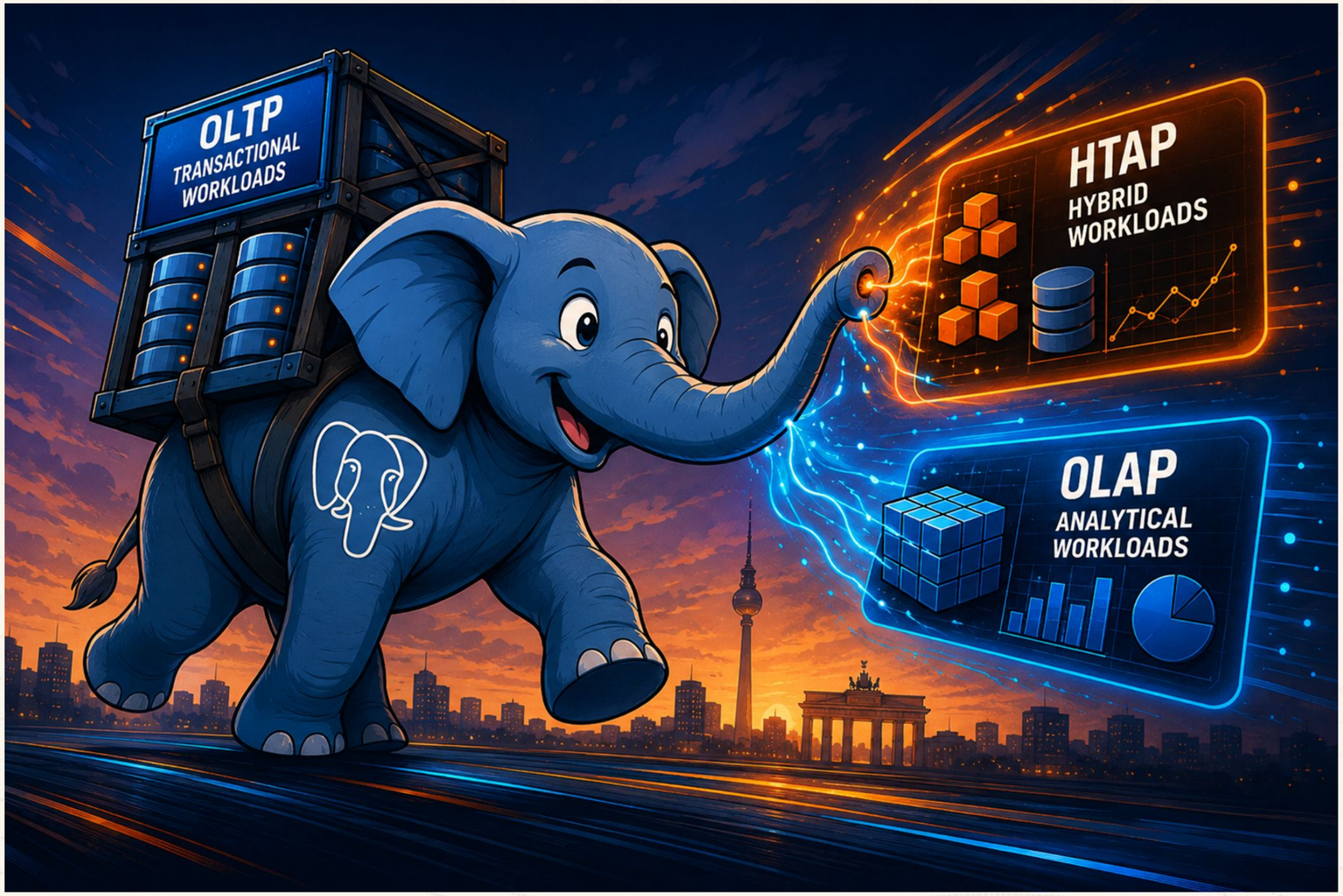
Co-Founder of benchANT



benchANT

- PhD in benchmarking cloud and database systems
- All about performance, databases and systems
- Enjoys to demystify the black art of database benchmarking
- Loves every kind of racket sports

From OLTP to OLAP: Is PostgreSQL Eating Analytics Too?



How Are Analytics Vendors Building on PostgreSQL?

Databricks Agrees to Acquire Neon to Deliver Serverless Postgres for Developers + AI Agents

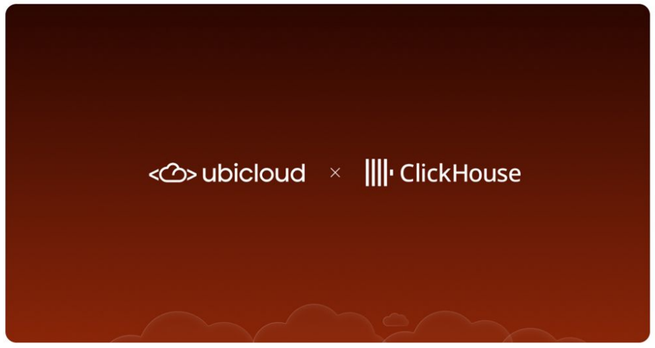
May 14, 2025



Mooncake Labs joins Databricks to accelerate the vision of Lakebase

Today we're excited to announce our partnership with ClickHouse: Managed Postgres as part of the ClickHouse Cloud platform, powered by Ubicloud Postgres. ClickHouse customers can now use Postgres for their transactional workloads, tightly integrated with ClickHouse for analytics.

This gives you a unified data stack: Postgres for transactions, ClickHouse for analytics, with native capabilities to sync and query data across datastores. Ubicloud Postgres, backed by fast, local NVMe storage, powers the Postgres experience.

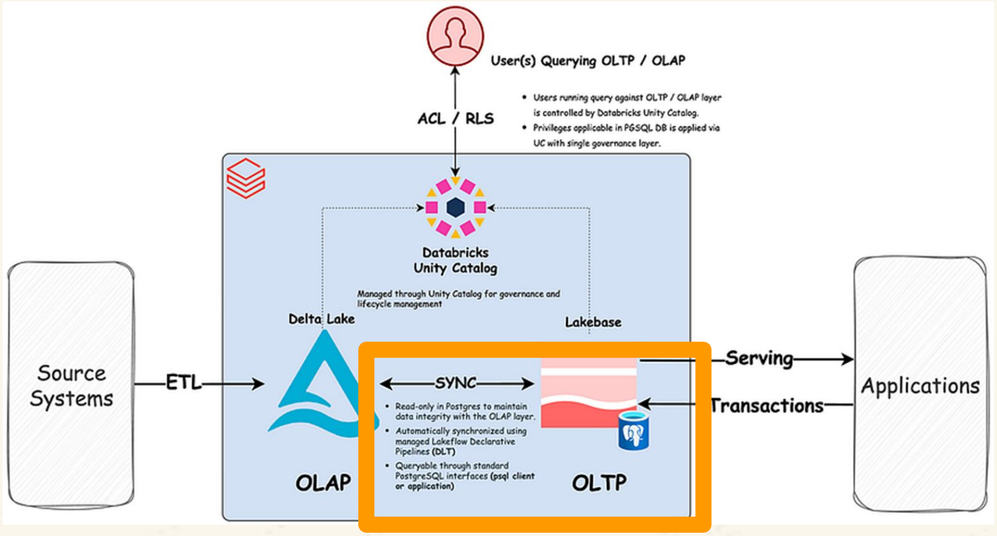


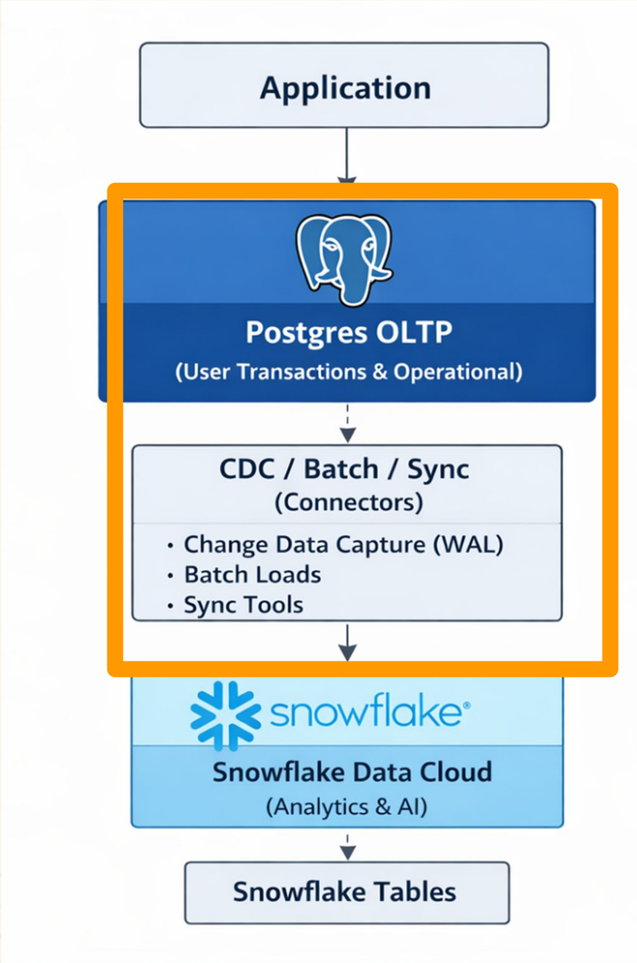


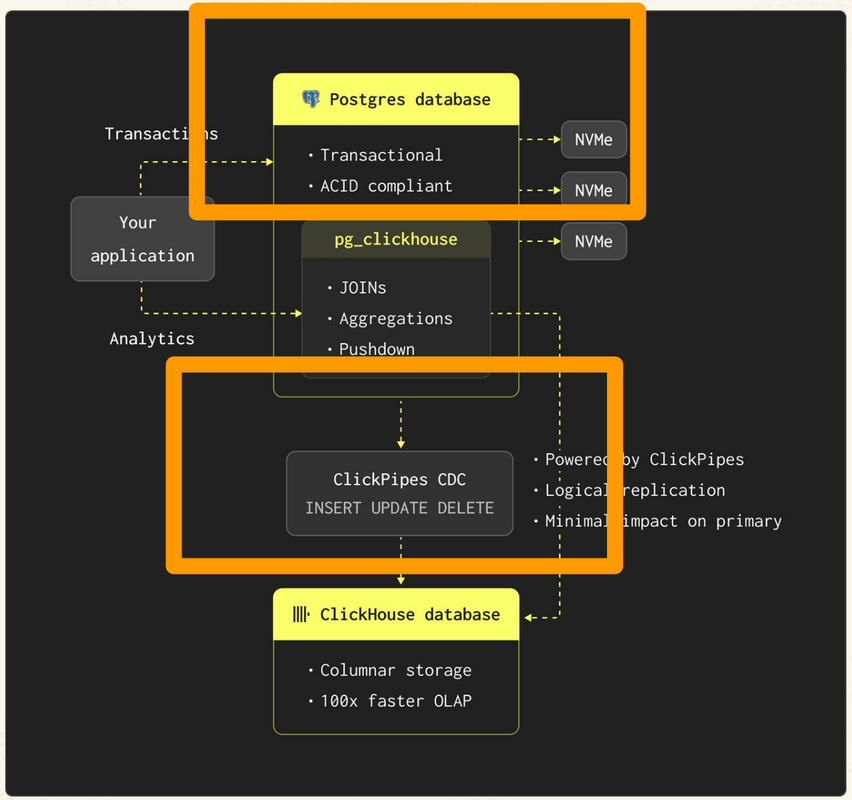


Snowflake Acquires Crunchy Data to Bring Enterprise Ready Postgres Offering to the AI Data Cloud

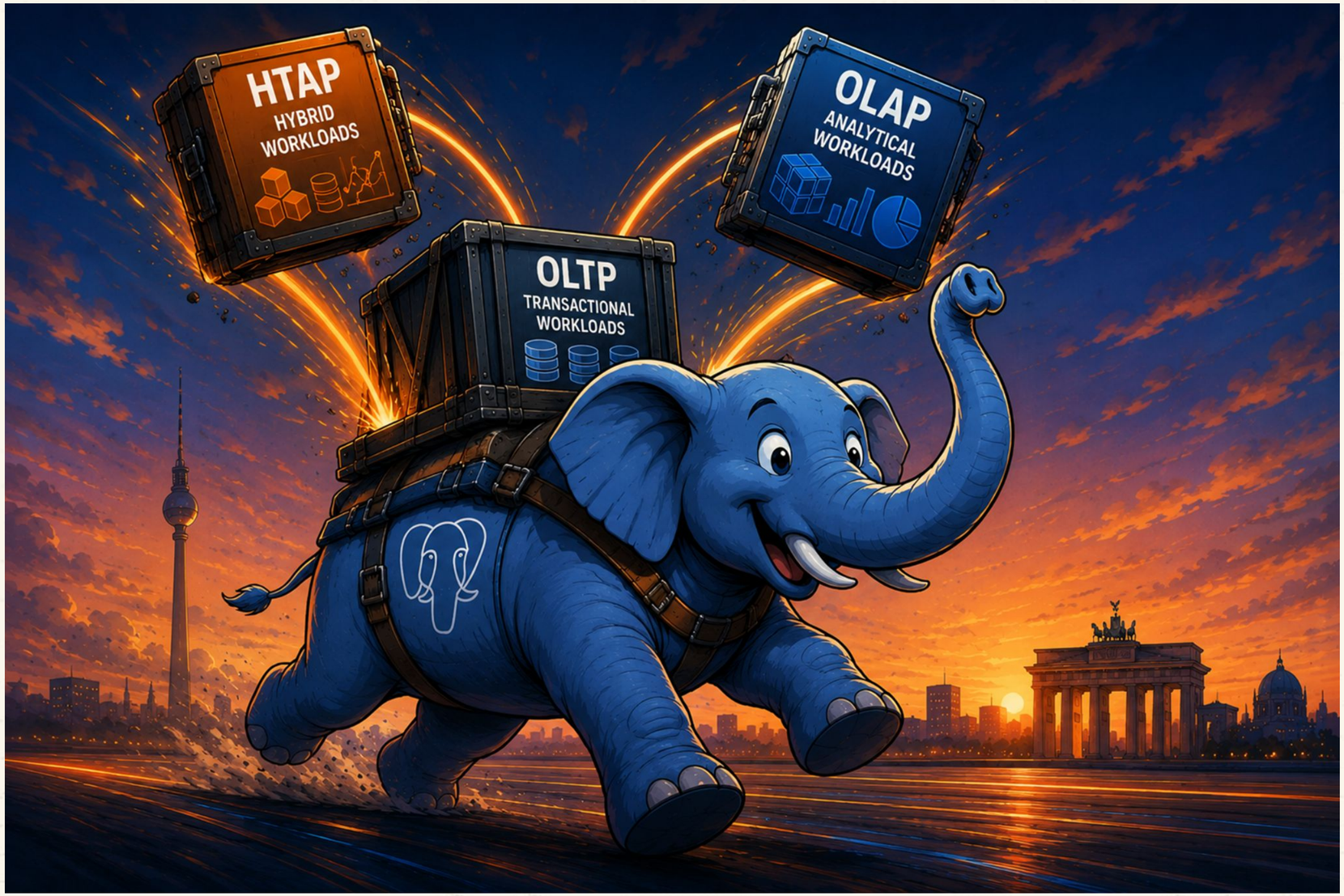
How Are Analytics Vendors Building on PostgreSQL?







Why Are Analytics Vendors Betting on PostgreSQL?



What Makes an Analytical Workload?

- Analytical domains: data warehouse, business intelligence, ad-hoc, dashboards, real-time analytics, search, agentic (?)
- Data set size: disk bound, 100 GB → PB
- Data schema: normalized vs. denormalized
- Query types: complex including joins, aggregations, grouping
- Data access: read-dominated with batched/streamed inserts, depending on the application domain (e.g. real-time analytics)

Can we turn PostgreSQL into an Analytical Database?

Embedding DuckDB in PostgreSQL via pg_duckdb

- DuckDB is a lightweight embedded database that provides a columnar and vectorized execution engine for analytical workloads
- pg_duckdb (3.1k GitHub stars) is maintained by MotherDuck Inc.
- pg_duckdb integrates DuckDB's columnar-vectorized analytics engine into PostgreSQL
- read/write data from/to data lakes and MotherDuck
- More details on https://github.com/duckdb/pg_duckdb

Embedding DuckDB into PostgreSQL

- Current release is v1.1.1 (released Dec 18, 2025)
- Execute analytics queries without code changes

```
SET duckdb.force_execution = true;
SELECT
    order_date,
    COUNT(*) AS number_of_orders,
    SUM(amount) AS total_revenue
FROM
    orders
GROUP BY
    order_date
ORDER BY
     order_date;
```

Distributed Columnar Storage with Citus

- Actively maintained PostgreSQL extension with > 12.5k GitHub stars (Citus Data acquired by Micorsoft in 2019)
- Citus primary focus is to turn PostgreSQL into a distributed database by adding distributed tables and a distributed query engine
- It also adds support for a columnar storage engine that compresses data, speeds up scans, and supports fast projections, both on regular and distributed
- Citus columnar is the successor of cstore_fdw
- More details on https://github.com/citusdata/citus/tree/main and https://github.com/citusdata/citus/tree/main/src/backend/columnar

Distributed Columnar Storage with Citus

- Current release is v14.2.0 (released Aug 6, 2026)
- Execute analytics queries without query code changes

```
# enable columnar engine for table
CREATE TABLE my_columnar_table
(
    id INT,
    i1 INT,
    n NUMERIC,
    t TEXT
 ) USING columnar;
 # configure columnar engine
 ALTER TABLE my_columnar_table SET
   (columnar.compression = none, columnar.stripe_row_limit = 10000);
 # enable partitioned columnar tables
 CREATE TABLE parent(ts timestamptz, i int, n numeric, s text)
   PARTITION BY RANGE (ts);
```

Benchmarking PostgreSQL Analytics

Benchmark Objectives

- Is there a silver bullet to push PostgreSQL analytics performance to the ClickHouse level?

- What is the performance improvement of pg_duckdb/Citus over native PostgreSQL?

Technical Benchmark Setup

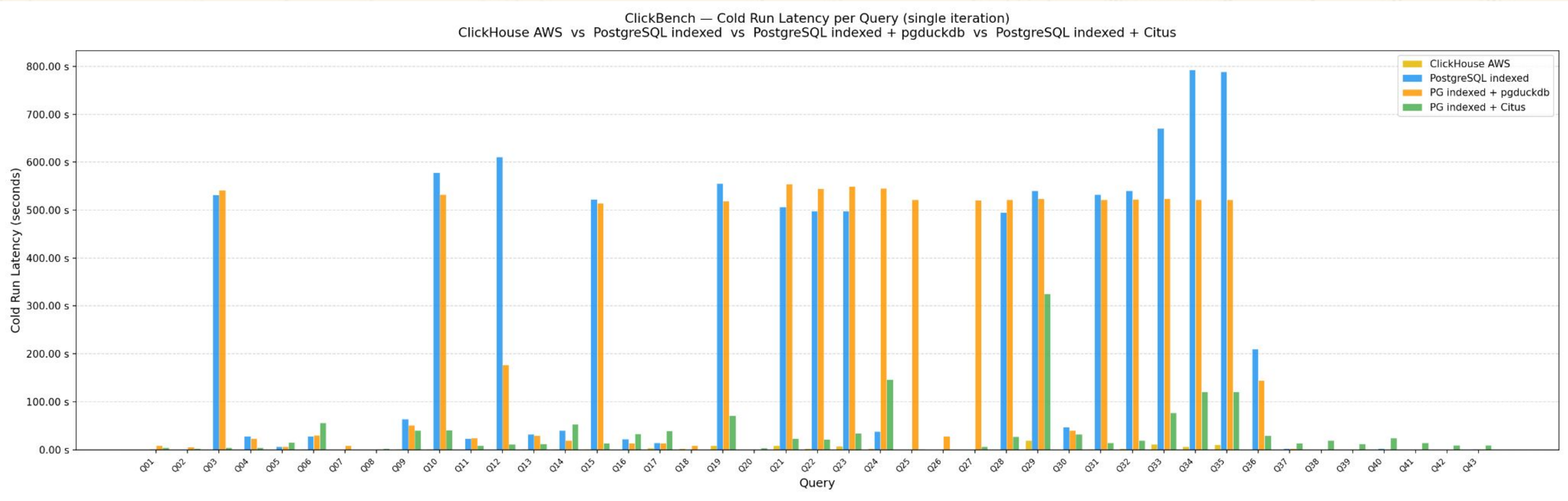
- EC2 m7i.2xlarge VM with 8 vCores/32GB RAM and 500GB GP3 storage
- PostgreSQL Version 17 with an analytical workload config by https://pgtune.leopard.in.ua/
  - pg_duckdb v1.1.1
  - Citus 14.0
- Clickhouse Cloud version 25.12 on AWS with 2 nodes of 4 vCores/16 GB RAM
- Benchmark instance is deployed on a c6i.4xlarge VM in the same region the PostgreSQL instance
- Full data set on GitHub: https://github.com/benchANT/p99-conf-26-postgresql-goes-columnar

ClickBench Results

ClickBench Setup

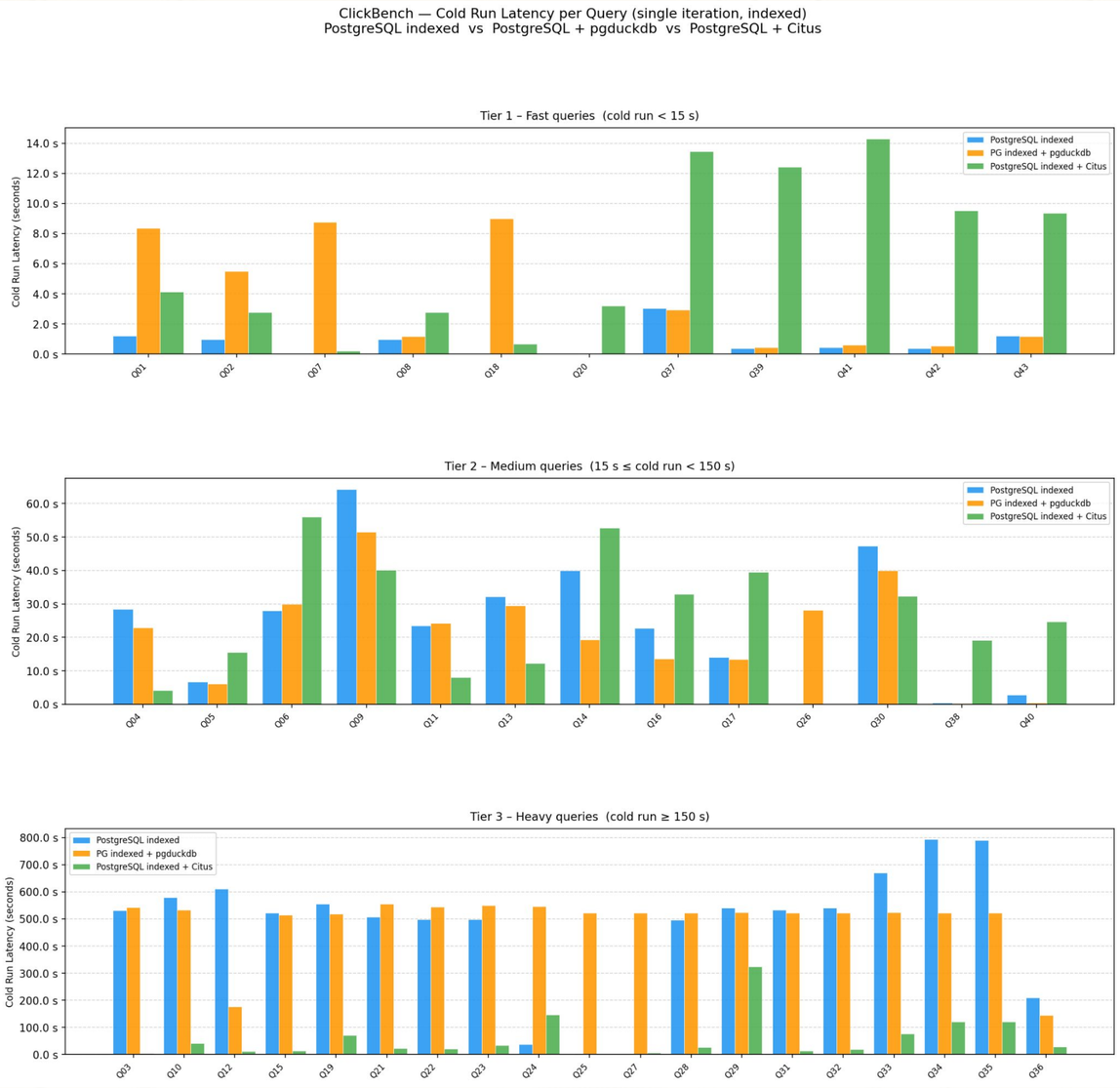
- 100M ClickBench data set from https://github.com/ClickHouse/ClickBench/tree/main
- Single table, 105 columns, 43 analytical queries
- Single threaded execution
- PostgreSQL,pg_duckdb and Citus schema, indexes and queries from https://github.com/ClickHouse/ClickBench/ (**Citus schema extended with PG indexes**)
- Benchmark execution and metric reporting is integrated into the benchANT platform but follows the ClickBench methodology

ClickBench Results



Where is ClickHouse?

ClickBench Results — Zoomed In



Query Tier	PostgreSQL	pg_duckdb	Citus
Wins	9	0	0
Draws	2		

Query Tier	PostgreSQL	pg_duckdb	Citus
Wins	1	4	5
Draws	3		

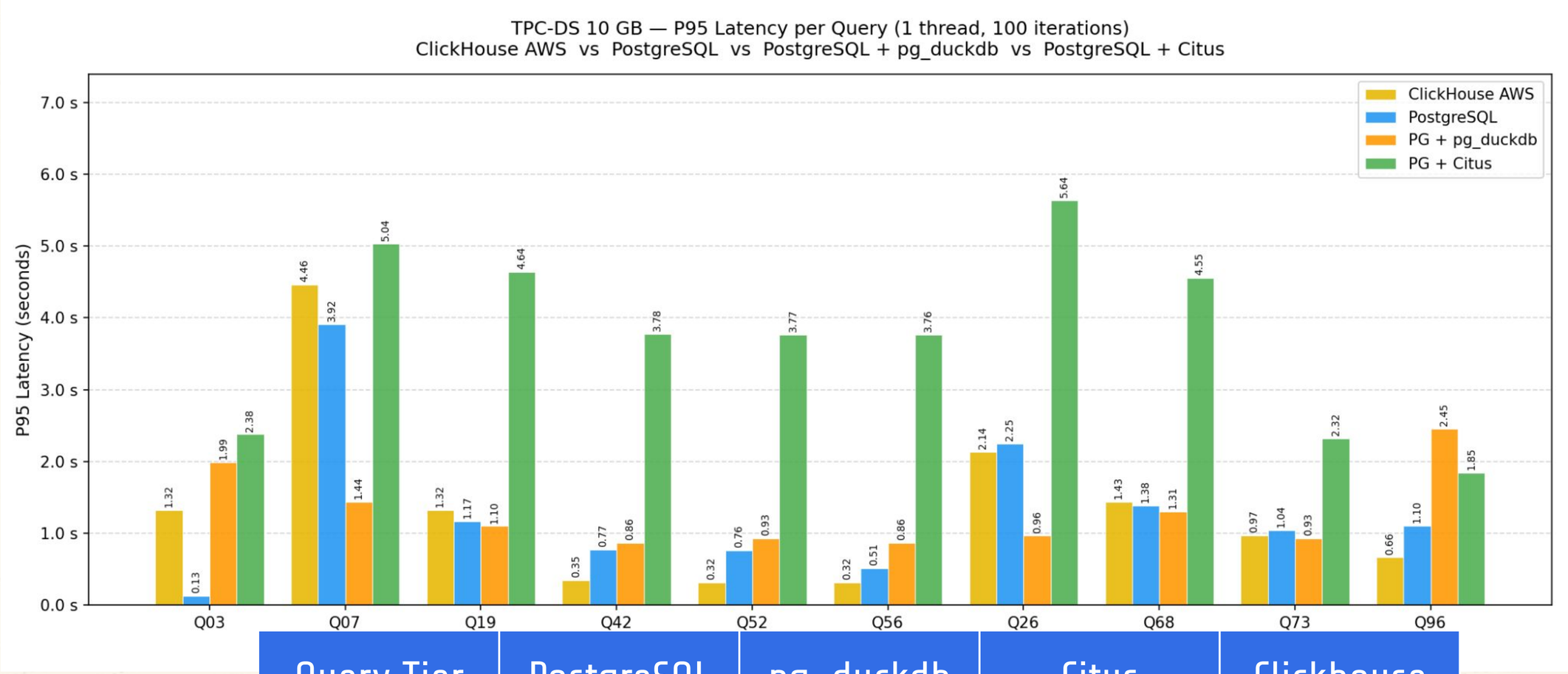
Query Tier	PostgreSQL	pg_duckdb	Citus
Wins	3	0	16
Draws	0		

benchANT TPC-DS Results

benchANT TPC-DS Setup

- Analytics domain: data warehouse
- Data set size is 10GB (SF=10)
- 24 tables in snowflake schema
- Picked 10 out of 99 official TPC-DS queries with multi-way joins, aggregations, and grouping
- Single threaded and sequential query execution
- 100 iterations per query
- Citus: columnar engine is only enabled for store_returns, inventory, catalog_returns, web_returns, web_sales, catalog_sales, store_sales

benchANT TPC-DS Results



Query Tier	PostgreSQL	pg_duckdb	Citus	Clickhouse
Wins	1	2	0	4
Draws		3		

What We Skipped (for Now)

- Native analytical PostgreSQL features (materialized view, cube, roll-up, …)
- Query execution plan analysis
- Larger ClickBench and TPC-DS data sets
- Concurrent query execution
- Data ingest performance
- Real-time analytic workloads (data freshness, ingest-vs-query, …)

Conclusion

- PostgreSQL is expanding beyond OLTP
- There is no single silver bullet "analytical PostgreSQL extension"
- pg_duckdb & Citus show promising but workload depending results
- benchANT’s benchmark road ahead
  - Native PostgreSQL features and more analytical extensions
  - Concurrent and real-time analytic workload shapes
  - PostgreSQL + external analytics engine for HTAP and OLAP

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Thank you! Let's connect.



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